

8. The number of red balls obtained will have the binomial distribution with parameters $n = 20$ and $p = 0.1$. The required probability can be found from the table of binomial probabilities in the back of the book. Add up the numbers in the $n = 20$ and $p = 0.1$ section from $k = 4$ to $k = 20$. Or add up the numbers from $k = 0$ to $k = 3$ and subtract the sum from 1. The answer is 0.1330.
9. We need $\sum_{x=0}^{\infty} f(x) = 1$, which means that $c = 1/\sum_{x=0}^{\infty} 2^{-x}$. The last sum is known from Calculus to equal $1/(1 - 1/2) = 2$, so $c = 1/2$.
10. (a) The p.f. of X is $f(x) = c(x+1)(8-x)$ for $x = 0, \dots, 7$ where c is chosen so that $\sum_{x=0}^7 f(x) = 1$. So, c is one over $\sum_{x=0}^7 (x+1)(8-x)$, which sum equals 120, so $c = 1/120$. That is $f(x) = (x+1)(8-x)/120$ for $x = 0, \dots, 7$.
- (b) $\Pr(X \geq 5) = [(5+1)(8-5) + (6+1)(8-6) + (7+1)(8-7)]/120 = 1/3$.
11. In order for the specified function to be a p.f., it must be the case that $\sum_{x=1}^{\infty} \frac{c}{x} = 1$ or equivalently $\sum_{x=1}^{\infty} \frac{1}{x} = \frac{1}{c}$. But $\sum_{x=1}^{\infty} \frac{1}{x} = \infty$, so there cannot be such a constant c .

3.2 Continuous Distributions

Commentary

This section ends with a brief discussion of probability distributions that are neither discrete nor continuous. Although such distributions have great theoretical interest and occasionally arise in practice, students can go a long way without actually concerning themselves about these distributions.

Solutions to Exercises

1. We compute $\Pr(X \leq 8/27)$ by integrating the p.d.f. from 0 to $8/27$.

$$\Pr\left(X \leq \frac{8}{27}\right) = \int_0^{8/27} \frac{2}{3}x^{-1/3}dx = x^{2/3}\Big|_0^{8/27} = \frac{4}{9}.$$

2. The p.d.f. has the appearance of Fig. S.3.1.

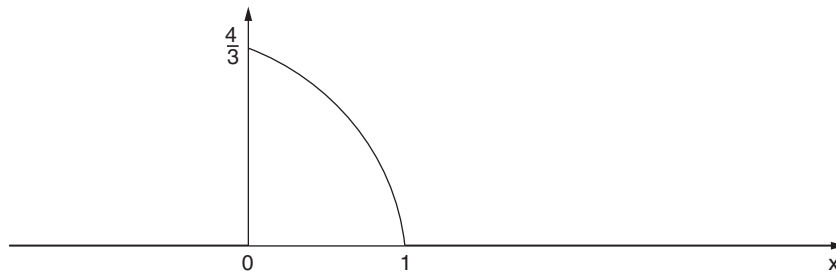


Figure S.3.1: Figure for Exercise 2 of Sec. 3.2.

(a) $\Pr\left(X < \frac{1}{2}\right) = \int_0^{1/2} 4(1-x^3)dx/3 = 0.6458.$

$$(b) \Pr\left(\frac{1}{4} < X < \frac{3}{4}\right) = \int_{1/4}^{3/4} 4(1-x^3)dx/3 = 0.5625.$$

$$(c) \Pr\left(X > \frac{1}{3}\right) = \int_{1/3}^1 4(1-x^3)dx/3 = 0.5597.$$

3. The p.d.f. has the appearance of Fig. S.3.2.

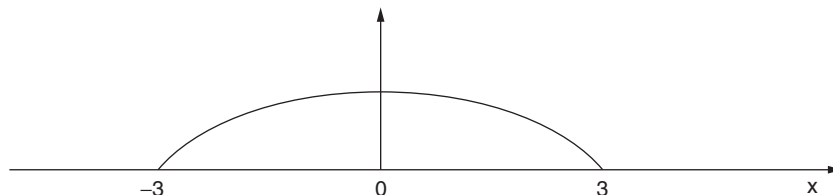


Figure S.3.2: Figure for Exercise 3 of Sec. 3.2.

$$(a) \Pr(X < 0) = \frac{1}{36} \int_{-3}^0 (9-x^2)dx = 0.5.$$

$$(b) \Pr(-1 < X < 1) = \frac{1}{36} \int_{-1}^1 (9-x^2)dx = 0.4815.$$

$$(c) \Pr(X > 2) = \frac{1}{36} \int_2^3 (9-x^2)dx = 0.07407.$$

The answer in part (a) could also be obtained directly from the fact that the p.d.f. is symmetric about the point $x = 0$. Therefore, the probability to the left of $x = 0$ and the probability to the right of $x = 0$ must each be equal to $1/2$.

4. (a) We must have

$$\int_{-\infty}^{\infty} f(x) dx = \int_1^2 cx^2 dx = \frac{7}{3}c = 1.$$

Therefore, $c = 3/7$. This p.d.f. has the appearance of Fig. S.3.3.

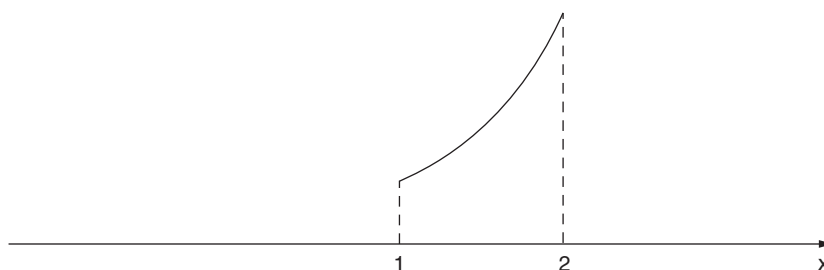


Figure S.3.3: Figure for Exercise 4a of Sec. 3.2.

$$(b) \int_{3/2}^2 f(x)dx = 37/56.$$

5. (a) $\int_0^t \frac{1}{8}x dx = 1/4$, or $t^2/16 = 1/4$. Hence, $t = 2$.

(b) $\int_t^4 (x/8) dx = 1/2$, or $1 - t^2/16 = 1/2$. Hence, $t = \sqrt{8}$.

6. The value of X must be between 0 and 4. We will have

$$Y = \begin{cases} 0 & \text{if } 0 \leq X < 1/2, \\ 1 & \text{if } 1/2 < X < 3/2, \\ 2 & \text{if } 3/2 < X < 5/2, \\ 3 & \text{if } 5/2 < X < 7/2, \\ 4 & \text{if } 7/2 < X \leq 4. \end{cases}$$

We need not worry about how to define Y if $X \in \{1/2, 3/2, 5/2, 7/2\}$, because the probability that X will be equal to one of these four values is 0. It now follows that

$$\begin{aligned} \Pr(Y = 0) &= \int_0^{1/2} f(x) dx = \frac{1}{64}, \\ \Pr(Y = 1) &= \int_{1/2}^{3/2} f(x) dx = \frac{1}{8}, \\ \Pr(Y = 2) &= \int_{3/2}^{5/2} f(x) dx = \frac{1}{4}, \\ \Pr(Y = 3) &= \int_{5/2}^{7/2} f(x) dx = \frac{3}{8}, \\ \Pr(Y = 4) &= \int_{7/2}^4 f(x) dx = \frac{15}{64}. \end{aligned}$$

7. Since the uniform distribution extends over an interval of length 10 units, the value of the p.d.f. must be $1/10$ throughout the interval. Hence,

$$\int_0^7 f(x) dx = \frac{7}{10}.$$

8. (a) We must have

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} c \exp(-2x) dx = \frac{1}{2}c = 1.$$

Therefore, $c = 2$. This p.d.f. has the appearance of Fig. S.3.4.

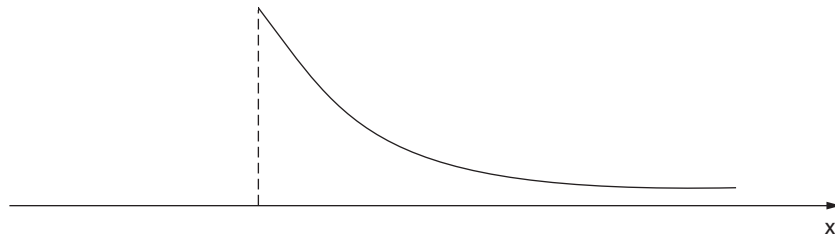


Figure S.3.4: Figure for Exercise 8a of Sec. 3.2.

$$(b) \int_1^2 f(x) dx = \exp(-2) - \exp(-4).$$

9. Since $\int_0^{\infty} 1/(1+x) dx = \infty$, there is no constant c such that $\int_0^{\infty} f(x) dx = 1$.

10. (a) We must have

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 \frac{c}{(1-x)^{1/2}} dx = 2c = 1.$$

Therefore $c = 1/2$. This p.d.f. has the appearance of Fig. S.3.5.

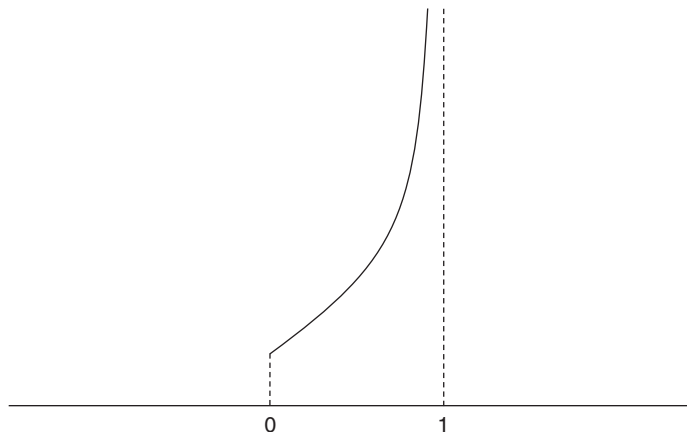


Figure S.3.5: Figure for Exercise 10a of Sec. 3.2.

It should be noted that although the values of $f(x)$ become arbitrarily large in the neighborhood of $x = 1$, the total area under the curve is equal to 1. It is seen, therefore, that the values of a p.d.f. can be greater than 1 and, in fact, can be arbitrarily large.

(b) $\int_0^{1/2} f(x) dx = 1 - (1/2)^{1/2}.$

11. Since $\int_0^1 (1/x) dx = \infty$, there is no constant c such that $\int_0^1 f(x) dx = 1$.

12. We shall find the c.d.f. of Y and evaluate it at 50. The c.d.f. of a random variable Y is $F(y) = \Pr(Y \leq y)$. In Fig. 3.1, on page 94 of the text, the event $\{Y \leq y\}$ has area $(y - 1) \times (200 - 4) = 196(y - 1)$ if $1 \leq y \leq 150$. We need to divide this by the area of the entire rectangle, 29,204. The c.d.f. of Y is then

$$F(y) = \begin{cases} 0 & \text{for } y < 1, \\ \frac{196(y-1)}{29204} & \text{for } 1 \leq y \leq 150, \\ 1 & \text{for } y > 150. \end{cases}$$

So, in particular, $\Pr(Y \leq 50) = 0.3289$.

13. We find $\Pr(X < 20) = \int_0^{20} cx dx = 200c$. Setting this equal to 0.9 yields $c = 0.0045$.

3.3 The Cumulative Distribution Function

Commentary

This section includes a discussion of quantile functions. These arise repeatedly in the construction of hypothesis tests and confidence intervals later in the book.